1. Clinical Question & Core Hypothesis

Clinical Question

The Ontologist framed a core question of why robust local immune contexture (Immunoscore/TILs) and/or high tumor mutational burden (MSI/dMMR) do not uniformly predict durable benefit from immune-stimulating therapies, and how non■invasive readouts (ctDNA) might resolve this discordance. 2) Tier 1 paths used: I built on (a) the Immunoscore/immune-contexture axis by operationalizing it as TIL density (imaging proxy of Immunoscore) because it is directly measurable on H&E/WSI, (b) the MSI/dMMR path as a genomic feature that modulates neoantigen load and immune engagement, and (c) the ctDNA/liquid-biopsy path as a dynamic, non■invasive readout of emerging immune■evasive clones. These three form a mechanistic chain: neoantigen load (MSI/dMMR) -> local immune infiltration (TIL density) -> selection/escape dynamics detectable by ctDNA. 3) Tier 2 support: SARIFA, PDIA2, pZVR and methylation markers (SDC2/TFPI2) provide orthogonal evidence that gene-expression/epigenetic programs and cross■disease protein biomarkers can mark high■risk or immune■modulating tumor states; they support the plausibility that tumor-intrinsic programs modulate how TILs translate into control. 4) Why PFI: Progression-free interval is the most sensitive clinical endpoint here for capturing early failure/escape after an immune stimulus and aligns with ctDNA kinetics as an early surrogate of recurrence; it therefore best tests whether immune infiltration + genomic context produce durable disease control. 5) Deliberate exclusions: I excluded broad cross-disease markers (PSA/PDIA2 used in prostate contexts) and higher-level labels (the umbrella term Immunoscore) from the formal hypothesis to keep it tightly focused and testable in TCGA-style data: I used TIL density as the specific imaging feature and limited anchors to MSI/dMMR and ctDNA. Other Tier2/Tier3 candidates (SARIFA, stool methylation, pZVR) are listed as secondary_modifiers rather than included in the core claim to avoid overfitting and preserve mechanistic clarity?

Working Hypothesis

Baseline tumor-infiltrating lymphocyte (TIL) density predicts improved progression-free interval (PFI) only in the context of MSI/dMMR-associated high neoantigen load; in MSI/dMMR tumors, sustained neoantigen exposure can promote T-cell dysfunction and rapid emergence of immune-evasive ctDNA-detectable subclones, limiting PFI despite high TIL density. Therefore, combining baseline TIL density with MSI/dMMR status and early on-treatment ctDNA kinetics will provide superior stratification of durable PFI than TIL density alone.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	MSI/dMMR, ctDNA
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

MSI/dMMR: MSI/dMMR path as a genomic feature that modulates neoantigen load and immune engagement, and (c) the ctDNA/liquid-biopsy path as a dynamic, non■invasive readout of emerging immune■evasive clones.

ctDNA: ctDNA kinetics as an early surrogate of recurrence; it therefore best tests whether immune infiltration + genomic context produce durable disease control.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	SARIFA	Supporting modifier from knowledge graph
Secondary	PDIA2	Supporting modifier from knowledge graph
Secondary	PSA	Supporting modifier from knowledge graph
Secondary	pZVR	Supporting modifier from knowledge graph
Secondary	SDC2/TFPI2 methylation	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- SARIFA: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- PDIA2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.